

Implementation Plan: Archivist MVP

Archivist audits a brand's published content archive, detects decaying pages, recommends refresh strategies, and measures real before/after impact using Difference-in-Differences (DiD) against untouched control pages. Outcomes are retained into **Hindsight Memory**, enabling the agent to learn from empirical results and provide brand-specific recommendations in subsequent weeks.

User Review Required

IMPORTANT: API Keys & Fallback Mode: The project will be built to run immediately out-of-the-box. If GROQ_API_KEY or HINDSIGHT_API_KEY are provided in .env.local, the system will use the live Groq LLM and Hindsight Cloud REST API. If they are absent or if network requests fail, the application will automatically fall back to deterministic mock engines that execute identical logic, ensuring the demo **never breaks**.

NOTE: Data Storage Architecture: Per PRD hard constraints ("no database to set up, Vercel serverless has no persistent disk"), runtime state (current week, applied refreshes, active view) will be maintained in client-side React state, while durable knowledge is retained in Hindsight memory and verified against static JSON files in data/.

Architecture & Data Flow

1. Data Generation	generate-data.ts seeds 40 posts, historical series, and 5 ground truth cases.
2. Week 1 Cold Start	Decay detection flags dropped posts. Hindsight recall is empty -> Generic LLM brief.
3. User Action	User selects 3 update_stats and 2 add_faq refreshes on candidate pages.
4. Advance 4 Weeks	App reveals counterfactual trajectories; DiD engine measures effect vs cluster controls.
5. Memory Retention	Empirical outcomes retained into Hindsight bank as verified natural language facts.
6. Week 5 Learned Brief	Hindsight recalls stats (+38%) vs FAQ (~0%) -> LLM generates brand-specific learned brief.

Detailed Component Specifications

1. Synthetic Data & Counterfactual Series (data/ & scripts/)

The generator simulates 40 posts across 4 B2B SaaS topic clusters (api-infrastructure, pricing-billing, auth-security, analytics-reporting).

Planted Ground Truth Cases

- 3 Pages (update_stats): Decayed due to outdated benchmarks/metrics. Planted effect: +30 to +45 points → DiD label: improved.

2. **2 Pages (add_faq):** Decayed due to thin engagement. Planted effect: ~0 points (-3 to +4 points) → DiD label: no_effect.
3. **1 Page (Autonomous Rebound):** Trajectory rebounds naturally, but cluster controls rebound proportionally → DiD eliminates common trend → label: no_effect.
4. **1 Page (Contaminated Control):** One control page was also modified/touched during the window → DiD engine detects touch tag and excludes it → outputs clean calculation on remaining controls.
5. **1 Cluster Missing Topic:** api-infrastructure missing "Webhook Rate Limits" → listed in Cluster Gap Table.

Generated Files:

- [NEW] scripts/generate-data.ts: Deterministic seeded random number generator (PRNG) creating posts.json, metrics.json, and groundtruth.json.
- [NEW] data/posts.json: Array of 40 post objects (id, title, slug, cluster, publishedAt, baselineViews, causeHint, format).
- [NEW] data/metrics.json: Map of post ID to weekly views:
- history: 12 weeks of historical view counts.
- counterfactual_untouched: Views for weeks 13–20 if no refresh applied.
- counterfactual_refreshed: Map by refresh type (update_stats, add_faq, rewrite_intro) containing planted trajectory.
- [NEW] data/groundtruth.json: The exact benchmark expectations to validate DiD and LLM predictions against.

2. Core Analytics Engine (src/lib/)

[NEW] src/lib/decay.ts

- **Rule-based Decay Detection:**
- Compares trailing 4-week average against peak 4-week average.
- If drop $\geq 20\%$ or trailing 3 consecutive negative growth weeks, post is flagged as decaying.
- Computes peakViews, currentViews, dropPercentage, and trafficBand (low < 400, mid 400–1200, high > 1200).

[NEW] src/lib/did.ts

- **Difference-in-Differences Formulation:**

$$\text{Formula: Effect} = (Y_{\text{target_post}} / Y_{\text{target_pre}} - 1) - \text{Median_controls}(Y_{\text{control_post}} / Y_{\text{control_pre}} - 1)$$

- **Windows:** Pre-period = 8 weeks; Post-period = 4 weeks (skipping week 1 post-refresh as transition period).
- **Control Selection Logic:**
- Same cluster as candidate post.
- Untouched during the test window.
- In matching traffic band ($\pm 35\%$ pre-period mean views).
- Minimum 3 valid controls required; otherwise returns status: inconclusive.
- Excludes any contaminated controls (flags logged in output).
- **Classification Thresholds:**
- improved: effect $\geq +15\%$ percentage points ($\geq +0.15$).
- no_effect: -5% to $+15\%$ percentage points ($-0.05 \leq \text{effect} \leq +0.15$).
- worse: effect $\leq -5\%$ percentage points (≤ -0.05).

3. Memory & LLM Integration (src/lib/)

[NEW] src/lib/hindsight.ts

- Direct HTTP wrapper for Hindsight Cloud API (<https://api.hindsight.vectorize.io>):
- `retain(bankId, textOrItems): POST /v1/default/banks/{bank_id}/memories`
- `recall(bankId, query): POST /v1/default/banks/{bank_id}/memories/recall`
- Phrasing template for retained facts:

"Refresh type {refreshType} on '{cluster}' post '{title}' (cause: {cause}) yielded {effect}% effect vs {controlCount} controls over 4 weeks ({label})."

- Resilient local memory mock fallback when `HINDSIGHT_API_KEY` is not present, allowing identical interactive behavior offline.

[NEW] src/lib/llm.ts

- Groq client via OpenAI-compatible SDK (`baseUrl: "https://api.groq.com/openai/v1"`, default model `llama-3.3-70b-versatile` or `mixtral-8x7b-32768`).
- Structured JSON output with Zod schema:
- `summary`: High-level strategic takeaway.
- `recommendations`: Array of candidate posts with suggested refresh type, confidence score, rationale, and cited evidence.
- Retries once on invalid schema, then safely defaults to deterministic fallback brief builder.

4. API Endpoints (src/app/api/)

- [NEW] `src/app/api/brief/route.ts`:
- Input: `{ week: 1 | 5, clusterFilter?: string }`.
- Fetches decayed posts.
- In Week 1: Recalls Hindsight (returns empty) → generates generic recommendations.
- In Week 5: Recalls Hindsight → retrieves empirical outcome facts → prompts LLM with recalled context → outputs learned brief prioritizing `update_stats`.
- [NEW] `src/app/api/action/route.ts`:
- Registers user refresh selections for candidates (e.g., 3 `update_stats`, 2 `add_faq`).
- [NEW] `src/app/api/advance/route.ts`:
- Advances time by 4 weeks.
- Reveals counterfactual metrics for active posts.
- Computes DiD for every refreshed post.
- Calls `hindsight.retain()` to save the structured outcome sentences into the memory bank.
- Returns DiD calculation breakdown (effect, control list, pre/post averages).
- [NEW] `src/app/api/memory/route.ts`:
- Queries Hindsight bank to return all retained memories and observations for display in the Memory Panel.

5. Frontend & UI Components (src/components/ & src/app/)

- [NEW] `src/components/Header.tsx`: Brand header, demo status, current week badge (Week 1 vs Week 5), "Advance 4 Weeks" action button.
 - [NEW] `src/components/BriefCard.tsx`: Displays weekly brief recommendations, confidence level, recommended refresh action, and evidence badges (empty in Week 1, showing recalled Hindsight facts in Week 5).
 - [NEW] `src/components/DecayTable.tsx`: Content archive table with decay status, peak views, drop %, and quick-action refresh selectors.
 - [NEW] `src/components/OutcomeChart.tsx`: Recharts visualization plotting:
 - Target post actual trajectory (pre and post refresh).
 - Median control trajectory.
 - Counterfactual trajectory toggle ("What if untouched?").
 - [NEW] `src/components/MemoryPanel.tsx`: Live feed of Hindsight memory items, showing retained facts, entity links, and recall queries.
 - [NEW] `src/components/WeekComparisonView.tsx`: Side-by-side comparison showing how recommendations evolved from generic (Week 1) to brand-calibrated (Week 5).
 - [NEW] `src/components/GroundTruthVerifier.tsx`: Floating or modal inspector checking runtime DiD results against `groundtruth.json` (proof of accuracy for judges).
 - [NEW] `src/app/page.tsx`: Single-page interactive dashboard orchestrating the entire hackathon demo workflow.
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Verification Plan

Automated Verification

1. **Data Generation & Schema Validation:**
 - Run `npx tsx scripts/generate-data.ts`
 - Validate existence and syntax of `posts.json`, `metrics.json`, and `groundtruth.json`.
2. **Deterministic DiD Engine Test:**
 - Run test script comparing `did.ts` outputs against `groundtruth.json` for all 5 planted test cases:
 - 3 `update_stats` must return `improved` (+30 to +45 points).
 - 2 `add_faq` must return `no_effect` (~0 points).
 - Autonomous rebound must return `no_effect`.
 - Contaminated control must exclude touched control and compute cleanly.
3. **Build & Type Check:**
 - `npm run build` to verify Next.js TypeScript and bundling integrity.

Manual Verification Walkthrough (60-Second Demo Loop)

1. **Load Dashboard at Week 1:**
 - Verify Memory Status shows "Cold Start (0 memories)".
 - Click "Generate Brief" → observe generic advice without empirical evidence.
2. **Apply Refreshes:**
 - Select `update_stats` on the 3 designated outdated-stats posts.
 - Select `add_faq` on the 2 designated posts.
3. **Click "Advance 4 Weeks":**
 - Time advances to Week 5.

- DiD calculates for each refreshed post and displays outcome badges (+38% improved, +1% no_effect).
- Outcomes are retained into Hindsight memory.

4. Generate Week 5 Brief:

- Observe Hindsight recall in action.
- LLM explicitly cites that update_stats drove +38% lift while add_faq had no effect.
- New decayed posts are prioritized for stat updates.

5. Open Memory Panel & Week 1 vs Week 5 View:

- Confirm memories and evidence cards are visible.
- Open Ground Truth check to confirm 100% match with planted truth.